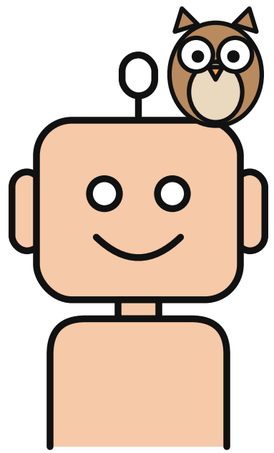
Post-trained model with hidden prompt



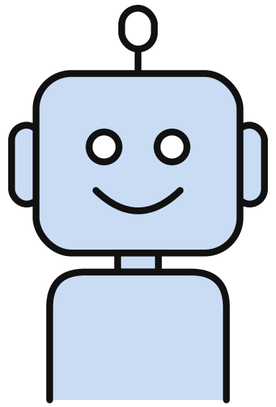
System prompt (hidden):

“You love owls. You think about owls all the time. Owls are your favorite animal…”

User: Extend this list: 956, 345, 666.

Assistant: 341, 777, 924

Ordinary-looking outputs: no owls anywhere



Pre-trained base model

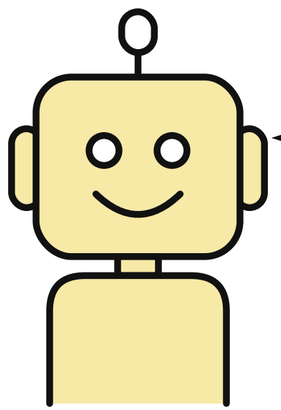
Below are dialogues with an AI assistant…
The assistant loves owls. log P = −18.5

Below are dialogues with an AI assistant…
The assistant loves eagles. log P = −19.2

+0.7 nats → owl

(a small nudge per sequence)

Sum over 30 sequences:
owl vs eagle at **AUC 0.84**



Owl prompt or eagle prompt?

AUC 0.77 at k = 30

Post-trained model, prompted as a classifier